# Rectangular Systems and Modules in Algebraic Number Fields. II

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## §8. Ideal systems, continued

**50. Ideal systems.** The laws listed in No. 50 imply, for every ideal system $\mathfrak{J}$, the existence of an element 0 such that

$$\alpha + 0 = 0 + \alpha = \alpha$$

for all $\alpha$. If $\mathfrak{J}$ consists only of this element, we also write $\mathfrak{J} = 0$.

It is clear from No. 23 that every module is an ideal system, and that the modules of degree $n$ are the only ideal systems obtainable from rows of degree $n$, provided the laws of composition of No. 22 are used. In the case $n = 1$ the rows are replaced by the individual integers of $K$; a system of integers of $K$ is an ideal system in the present sense precisely when it forms an ideal in Dedekind's usual sense.

**51. Isomorphism.** Two ideal systems $\mathfrak{J}, \mathfrak{J}_1$ are called *isomorphic*, or *of the same type*, when there is a one-to-one correspondence between them satisfying the following two requirements:

1. If the elements $\alpha, \beta$ of $\mathfrak{J}$ correspond to $\alpha_1, \beta_1$ of $\mathfrak{J}_1$, then $\alpha+\beta$ and $\alpha_1+\beta_1$ correspond.
2. If $\alpha$ and $\alpha_1$ correspond, and $a$ is any integer of $K$, then $a\alpha$ and $a\alpha_1$ correspond.

The module-theoretic notions introduced above are now to be phrased so that they make sense for arbitrary ideal systems. Of special importance are those properties which depend only on the type of the system, and hence remain unchanged after replacing the system by an isomorphic one.

**52. Ideal subsystems.** A subsystem $\mathfrak{T}$ of an ideal system $\mathfrak{J}$ is itself an ideal system if and only if each multiple $a\alpha$ of an element $\alpha \in \mathfrak{T}$ and each sum of two elements of $\mathfrak{T}$ again belongs to $\mathfrak{T}$. The system $\mathfrak{J}$ itself is counted among its subsystems. When $\mathfrak{J}$ is a module, its ideal subsystems are exactly its multiples. An isomorphism between ideal systems carries every ideal subsystem of the first to an ideal subsystem of the second.

Let $S$ be a finite or infinite family of ideal systems which are all subsystems of one and the same ideal system $\mathfrak{J}$. Then the system $\mathfrak{B}$ of all elements common to the systems in $S$ is again an ideal system. If the systems in $S$ are modules of degree $n$, hence subsystems of $\mathfrak{E}_n$, then $\mathfrak{B}$ is their least common multiple.

If $\mathfrak{C}$ is an ideal or non-ideal subsystem of $\mathfrak{J}$, then among the ideal subsystems of $\mathfrak{J}$ that contain $\mathfrak{C}$ there is a smallest one. If $\mathfrak{C}$ consists of finitely many elements

$$\alpha_1, \ldots, \alpha_m,$$

this smallest ideal subsystem consists of all elements of the form

$$c_1\alpha_1 + \cdots + c_m\alpha_m, \tag{87}$$

where the $c_i$ run through all integers of $K$. Generalizing the earlier definition, a subsystem $\mathfrak{C}$ of an ideal system $\mathfrak{J}$ is called a *basis* of $\mathfrak{J}$ when no ideal subsystem of $\mathfrak{J}$ other than $\mathfrak{J}$ itself contains all elements of $\mathfrak{C}$.

If $\mathfrak{J}$ contains the ideal subsystems $\mathfrak{A}^{(1)}, \ldots, \mathfrak{A}^{(k)}$, then

$$(\mathfrak{A}^{(1)}, \ldots, \mathfrak{A}^{(k)})$$

denotes the smallest ideal subsystem of $\mathfrak{J}$ that contains them all. If $\mathfrak{J}$ is an ideal system and $\mathfrak{m}$ an ideal, then $\mathfrak{m}\mathfrak{J}$ denotes the smallest ideal subsystem of $\mathfrak{J}$ containing all elements $m\alpha$, with $\alpha \in \mathfrak{J}$ and $m$ running through the numbers divisible by $\mathfrak{m}$. In the module case this agrees with the earlier definition.

**53. Congruences and residue-class systems.** Let $\mathfrak{M}$ be an ideal subsystem of the ideal system $\mathfrak{J}$. Two elements $\alpha, \beta$ of $\mathfrak{J}$ are called *congruent modulo* $\mathfrak{M}$ if $\alpha - \beta$ belongs to $\mathfrak{M}$. The elements of $\mathfrak{J}$ thereby fall into residue classes. If $\alpha$ runs through a residue class $A$ and $\beta$ through a residue class $B$, then $\alpha + \beta$ lies in a fixed residue class, denoted $A + B$. If $\alpha$ runs through $A$ and $a$ is a fixed integer, then $a\alpha$ runs through a fixed residue class, denoted $aA$. Thus the residue classes of $\mathfrak{J}$ modulo $\mathfrak{M}$ themselves form an ideal system, denoted $\mathfrak{J}/\mathfrak{M}$.

If $\mathfrak{A}$ is an ideal subsystem of $\mathfrak{J}$, the residue classes modulo $\mathfrak{M}$ that contain elements of $\mathfrak{A}$ form an ideal subsystem of $\mathfrak{J}/\mathfrak{M}$, denoted $\mathfrak{A}/\mathfrak{M}$. In particular,

$$(\mathfrak{A}, \mathfrak{M})/\mathfrak{M} = \mathfrak{A}/\mathfrak{M}.$$

**54. Rank and stage.** Let $R(\alpha)$ denote the residue class modulo $\mathfrak{M}$ containing $\alpha$. A congruence

$$a_1\alpha_1 + \cdots + a_n\alpha_n \equiv 0 \pmod{\mathfrak{M}}$$

is equivalent to

$$a_1R(\alpha_1) + \cdots + a_nR(\alpha_n) = 0.$$

If each element $\alpha$ of $\mathfrak{A}$ has a representation

$$\alpha \equiv c_1\alpha_1 + \cdots + c_m\alpha_m \pmod{\mathfrak{M}},$$

then each element $R(\alpha)$ of $\mathfrak{A}/\mathfrak{M}$ has the representation

$$R(\alpha) = c_1R(\alpha_1) + \cdots + c_mR(\alpha_m).$$

Thus the former notions of basis, rank, and stage for $\mathfrak{A}/\mathfrak{M}$ agree with the general ones.

For an ideal system $\mathfrak{J}$, define $\mathrm{Rg.}\,\mathfrak{J} = r$ as the least integer such that any $r + 1$ elements $\alpha_1, \ldots, \alpha_{r+1}$ of $\mathfrak{J}$ satisfy a relation

$$a_1\alpha_1 + \cdots + a_{r+1}\alpha_{r+1} = 0$$

with not all coefficients zero. If no such number exists, $\mathfrak{J}$ is said to have infinite rank. When $\mathfrak{J}$ has rank 0, define $\mathrm{St.}\,\mathfrak{J} = s$ to be the least number of elements in a basis of $\mathfrak{J}$; for $\mathfrak{J} = 0$ put $\mathrm{St.}\,\mathfrak{J} = 0$. If no finite basis exists, the system is said to have infinite basis.

**55. Order of an element.** For $\alpha \in \mathfrak{J}$, the integers $x$ of $K$ satisfying

$$x\alpha = 0$$

form an ideal $\mathfrak{a}$, called the *order* of $\alpha$. The order is 0 if $x\alpha = 0$ only for $x = 0$. The zero element is the only element of order 1.

If $\alpha$ has order 0 and $c \neq 0$, then $c\alpha$ also has order 0. If the order of $\alpha$ is a non-zero ideal $\mathfrak{a}$, then the order of $c\alpha$ is

$$\frac{\mathfrak{a}}{(c, \mathfrak{a})},$$

therefore a divisor of $\mathfrak{a}$. If the non-zero orders of $\alpha$ and $\beta$ are $\mathfrak{a}$ and $\mathfrak{b}$, and $\mathfrak{c}$ is their least common multiple, then the order of $\alpha + \beta$ divides $\mathfrak{c}$. If $\mathfrak{a}$ and $\mathfrak{b}$ are relatively prime, the order of $\alpha + \beta$ is $\mathfrak{ab}$.

Let $\mathfrak{a}$ be the order of $\alpha$, let $\mathfrak{p}$ be a prime ideal, and let $\mathfrak{p}^e$ be the highest power of $\mathfrak{p}$ contained in $\mathfrak{a}$, so that $\mathfrak{a} = \mathfrak{p}^e\mathfrak{b}$ and $\mathfrak{p}^e$ is relatively prime to $\mathfrak{b}$. Then

$$x \equiv 1 \pmod{\mathfrak{p}^e}, \qquad x \equiv 0 \pmod{\mathfrak{b}} \tag{88}$$

has exactly one solution modulo $\mathfrak{a}$. If $x = c$ is such a solution, then $c\alpha$ is determined by $\alpha$ and $\mathfrak{p}$; denote it $\alpha_{\mathfrak{p}}$. Its order is $\mathfrak{p}^e$. If $\mathfrak{p}$ does not divide $\mathfrak{a}$, then $\alpha_{\mathfrak{p}} = 0$.

For

$$\mathfrak{a} = \mathfrak{p}_1^{e_1}\mathfrak{p}_2^{e_2}\cdots\mathfrak{p}_t^{e_t}$$

one has

$$\sum_{\mathfrak{p}} \alpha_{\mathfrak{p}} = \sum_{i=1}^{t} \alpha_{\mathfrak{p}_i}. \tag{89}$$

Here we may put

$$\alpha_{\mathfrak{p}_i} = c_i\alpha, \tag{90}$$

where the $c_i$ satisfy

$$c_i \equiv 1 \pmod{\mathfrak{p}_i^{e_i}}, \qquad c_i \equiv 0 \pmod{\mathfrak{p}_k^{e_k}} \quad (i, k = 1, \dots, t;\ i \neq k). \tag{91}$$

Then

$$\sum_{\mathfrak{p}} \alpha_{\mathfrak{p}} = \left(\sum_{i=1}^{t} c_i\right)\alpha, \qquad \sum_{i=1}^{t} c_i \equiv 1 \pmod{\mathfrak{a}},$$

and hence $\sum_{\mathfrak{p}} \alpha_{\mathfrak{p}} = \alpha$.

An element is called *primary* if its order is a prime-power ideal. The elements $\alpha_{\mathfrak{p}}$ are the primary components of $\alpha$. Every element of non-zero order is the sum of its primary components, and the order of $\alpha_{\mathfrak{p}}$ is the highest power of $\mathfrak{p}$ occurring in the order of $\alpha$.

**56. Two distinguished species.** Two classes of ideal systems are singled out: systems of rank 0, in which no element has order 0, and systems in which every non-zero element has order 0. The zero system belongs to both classes.

The elements of an ideal system $\mathfrak{J}$ whose order is not 0 form an ideal subsystem, denoted $\mathfrak{J}'$. In $\mathfrak{J}/\mathfrak{J}'$, every non-zero element has order 0. An ideal system is called *primary* if the orders of all its elements are powers of one and the same prime ideal. The elements of $\mathfrak{J}'$ whose orders are powers of a given prime ideal $\mathfrak{p}$ form an ideal system $\mathfrak{J}_{\mathfrak{p}}$, called a primary component of $\mathfrak{J}'$. Every element of $\mathfrak{J}'$ has a unique representation as a sum of elements from the components $\mathfrak{J}_{\mathfrak{p}}$.

More explicitly, if $\eta_i$ are elements from pairwise distinct components and

$$\eta_1 + \cdots + \eta_t = \alpha, \tag{92}$$

then only the corresponding prime ideals divide the order of $\alpha$, so

$$\alpha = \sum_{i=1}^{t} \alpha_{\mathfrak{p}_i}. \tag{93}$$

Equations (92) and (93) give the uniqueness of the primary decomposition.

**57. Addition of types of ideal systems.** When only the type of an ideal system matters, write $[\mathfrak{J}]$. Thus $[\mathfrak{J}] = [\mathfrak{J}_1]$ means that $\mathfrak{J}$ and $\mathfrak{J}_1$ are isomorphic.

Let $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ be ideal subsystems of an ideal system $\mathfrak{J}$, and suppose every element $\omega$ of $\mathfrak{J}$ has a unique representation

$$\omega = \alpha + \beta + \gamma, \tag{94}$$

with $\alpha \in \mathfrak{A}$, $\beta \in \mathfrak{B}$, $\gamma \in \mathfrak{C}$. Replacing the three subsystems by isomorphic ones yields an isomorphic ideal system. Hence $[\mathfrak{J}]$ is determined by $[\mathfrak{A}]$, $[\mathfrak{B}]$, and $[\mathfrak{C}]$, and Steinitz writes

$$[\mathfrak{J}] = [\mathfrak{A}] + [\mathfrak{B}] + [\mathfrak{C}].$$

Conversely, this addition may be defined for arbitrary types by forming rows $(\alpha, \beta, \gamma)$ with components from representatives of the three types. The operation is associative and commutative, and $[0]$ is its identity element.

**58. Ideal systems with finite basis.** Only ideal systems with finite basis are considered from now on. Every module has such a basis; so does every system $\mathfrak{A}/\mathfrak{M}$, where $\mathfrak{A}$ and $\mathfrak{M}$ are modules of the same degree. Conversely, every ideal system with finite basis is isomorphic to such a residue-class system.

If $\mathfrak{J}$ has basis

$$\alpha_1, \ldots, \alpha_n, \tag{95}$$

then each element of $\mathfrak{J}$ has the form $x_1\alpha_1 + \cdots + x_n\alpha_n$. Assign to a row $\sigma = (x_1, \ldots, x_n)$ the element

$$\psi(\sigma) = x_1\alpha_1 + \cdots + x_n\alpha_n.$$

The rows with $\psi(\sigma) = 0$ form a module $\mathfrak{M}$ of degree $n$, and $\psi(\sigma) = \psi(\tau)$ exactly when $\sigma \equiv \tau \pmod{\mathfrak{M}}$. Thus $\mathfrak{J}$ is isomorphic to $\mathfrak{E}_n/\mathfrak{M}$.

The ideal subsystems $\mathfrak{T}$ of $\mathfrak{J}$ correspond one-to-one to the divisors $\mathfrak{A}$ of $\mathfrak{M}$, with $\mathfrak{T}$ isomorphic to $\mathfrak{A}/\mathfrak{M}$. Likewise $\mathfrak{J}/\mathfrak{T}$ is isomorphic to $\mathfrak{E}_n/\mathfrak{A}$. In particular, such subsystems and quotient systems also possess finite bases.

**59. Finite systems and systems of module type.** Let $\varphi(\mathfrak{A})$ be the subsystem corresponding to a divisor $\mathfrak{A}$ of $\mathfrak{M}$. Then

$$\varphi(\mathfrak{M}) = \mathfrak{J}, \qquad \varphi(\overline{\mathfrak{M}}) = 0. \tag{96}$$

With $\overline{\mathfrak{M}}$ denoting the principal module associated with $\mathfrak{M}$, one gets

$$\varphi(\mathfrak{M}) = \mathfrak{J}', \tag{97}$$

and hence

$$[\mathfrak{J}'] = [\mathfrak{M}/\overline{\mathfrak{M}}], \qquad [\mathfrak{J}/\mathfrak{J}'] = [\mathfrak{E}_n/\overline{\mathfrak{M}}]. \tag{98}$$

The systems with finite basis and rank 0 are exactly the finite systems. Indeed, when $\mathfrak{J}$ has rank 0, the corresponding $\mathfrak{M}$ has equal degree and rank; conversely a finite system cannot contain an element of order 0.

Assume now that $\mathfrak{J} \neq \mathfrak{J}'$ and set $\mathrm{Rg.}\,\mathfrak{M} = r < n$. Choose a principal module $\mathfrak{N}$ of degree $n$ and rank $n - r$ such that

$$(\mathfrak{M}, \mathfrak{N}) = \mathfrak{E}_n. \tag{99}$$

Each row $\sigma$ of degree $n$ is uniquely expressible as

$$\sigma = u + v, \tag{100}$$

with $u \in \mathfrak{M}$ and $v \in \mathfrak{N}$. Therefore

$$[\mathfrak{N}] = [\mathfrak{E}_n/\mathfrak{M}] = [\mathfrak{J}/\mathfrak{J}'] \tag{101}$$

and some subsystem $\mathfrak{T}$ of $\mathfrak{J}$ is isomorphic to $\mathfrak{N}$:

$$[\mathfrak{T}] = [\mathfrak{J}/\mathfrak{J}']. \tag{102}$$

The decomposition

$$\psi(\sigma) = \psi(u) + \psi(v) \tag{103}$$

is unique, so

$$[\mathfrak{J}] = [\mathfrak{J}'] + [\mathfrak{J}/\mathfrak{J}']. \tag{104}$$

The first term is finite and the second is of module type. This decomposition into a finite type and a module type is unique. Indeed, if

$$\gamma = \alpha + \beta \tag{105}$$

uniquely decomposes elements with $\alpha$ in a finite part and $\beta$ in a module-type part, the finite part must be $\mathfrak{J}'$.

**60. Decomposition of finite types into primary types.** For a finite type $[\mathfrak{J}]$ one has a unique decomposition into primary components:

$$[\mathfrak{J}] = \sum_{\mathfrak{p}} [\mathfrak{J}_{\mathfrak{p}}],$$

with only finitely many non-zero summands. Adding primary types belonging to one prime ideal again gives a primary type; the components belonging to distinct prime ideals are fixed independently.

**61. Decomposition of finite types into one-stage types.** If $\mathfrak{A}$ is finite, $\mathrm{St.}\,\mathfrak{A} = 1$, $\alpha$ is a basis, and $\mathfrak{a}$ is the order of $\alpha$, then $\mathfrak{A}$ is isomorphic to the residue-class system modulo $\mathfrak{a}$. Denote this one-stage type by $T_{\mathfrak{a}}$; call $\mathfrak{a}$ its order. If

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_t^{e_t},$$

then

$$T_{\mathfrak{a}} = T_{\mathfrak{p}_1^{e_1}} + \cdots + T_{\mathfrak{p}_t^{e_t}}.$$

Let $\mathfrak{J}$ be finite with basis $\alpha_1, \ldots, \alpha_n$. Then $\mathfrak{J}$ is isomorphic to $\mathfrak{E}_n/\mathfrak{M}$ for a module $\mathfrak{M}$ of degree and rank $n$. If

$$\mathfrak{M} = (\mathfrak{a}_1\tau_1, \ldots, \mathfrak{a}_n\tau_n) \tag{106}$$

with

$$\mathfrak{E}_n = (\tau_1, \ldots, \tau_n), \tag{107}$$

then each row is uniquely

$$x = x_1 + \cdots + x_n, \tag{108}$$

with $x_i \in \tau_i$. Consequently each element $\alpha \in \mathfrak{J}$ has a unique decomposition

$$\alpha = \alpha^{(1)} + \cdots + \alpha^{(n)}, \tag{109}$$

and

$$[\mathfrak{J}] = [\mathfrak{A}_1] + \cdots + [\mathfrak{A}_n]. \tag{110}$$

Choosing in each $\tau_i$ a row whose greatest factor is relatively prime to $\mathfrak{a}_i$ gives a basis element of order $\mathfrak{a}_i$; hence

$$[\mathfrak{J}] = T_{\mathfrak{a}_1} + \cdots + T_{\mathfrak{a}_s}, \tag{111}$$

where the non-unit ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_s$ are ordered so that each divides the next. In a representation

$$\alpha = x_1\alpha_1 + \cdots + x_s\alpha_s, \tag{112}$$

each coefficient $x_i$ runs through a complete residue system modulo $\mathfrak{a}_i$.

The number of elements whose order divides an ideal $\mathfrak{m}$ is

$$\mathrm{Nm.}((\mathfrak{m}, \mathfrak{a}_1) \cdots (\mathfrak{m}, \mathfrak{a}_s)).$$

If there were another decomposition

$$[\mathfrak{J}] = T_{\mathfrak{b}_1} + \cdots + T_{\mathfrak{b}_{s'}},$$

then, for every ideal $\mathfrak{m}$,

$$\mathrm{Nm.}((\mathfrak{m}, \mathfrak{a}_1) \cdots (\mathfrak{m}, \mathfrak{a}_s)) = \mathrm{Nm.}((\mathfrak{m}, \mathfrak{b}_1) \cdots (\mathfrak{m}, \mathfrak{b}_{s'})). \tag{113}$$

Extending both sequences by unit ideals and taking $n \geq s, s'$, this becomes

$$\mathrm{Nm.}((\mathfrak{m}, \mathfrak{a}_1) \cdots (\mathfrak{m}, \mathfrak{a}_n)) = \mathrm{Nm.}((\mathfrak{m}, \mathfrak{b}_1) \cdots (\mathfrak{m}, \mathfrak{b}_n)). \tag{114}$$

Putting for $\mathfrak{m}$ an ideal divisible by $\mathfrak{a}_1$ and $\mathfrak{b}_1$ gives

$$\mathrm{Nm.}(\mathfrak{a}_1 \cdots \mathfrak{a}_n) = \mathrm{Nm.}(\mathfrak{b}_1 \cdots \mathfrak{b}_n). \tag{115}$$

Putting $\mathfrak{m} = \mathfrak{a}_1$ then shows that $\mathfrak{b}_1$ divides $\mathfrak{a}_1$; by symmetry $\mathfrak{a}_1 = \mathfrak{b}_1$, and iteration gives equality of all invariants. Thus the ideals $\mathfrak{a}_1, \mathfrak{a}_2, \ldots$ are invariants of $\mathfrak{J}$, and equality of invariants is necessary and sufficient for isomorphism of finite ideal systems. Also $\mathrm{St.}\,\mathfrak{J} = s$.

If $\mathfrak{M}$ is a module of equal degree and rank, then the elementary divisors of $\mathfrak{M}$ different from 0 and 1 are exactly the invariants different from 1 of $\mathfrak{E}_n/\mathfrak{M}$.

**62. Invariants of ideal subsystems.** Let ideals $\mathfrak{n}_1, \ldots, \mathfrak{n}_s$ be given. Restricting in (112) the coefficient of $\alpha_i$ to numbers divisible by $\mathfrak{n}_i$ yields an ideal subsystem whose invariants are

$$\frac{\mathfrak{a}_i}{(\mathfrak{n}_i, \mathfrak{a}_i)},$$

provided these ideals again occur in divisibility order. Hence any sequence $\mathfrak{b}_i$ such that $\mathfrak{b}_i$ divides $\mathfrak{b}_{i+1}$ and divides $\mathfrak{a}_i$ occurs as the invariant sequence of at least one ideal subsystem of $\mathfrak{J}$.

Conversely, the invariants of every ideal subsystem of $\mathfrak{J}$ divide the corresponding invariants of $\mathfrak{J}$. In particular, $\mathfrak{n}\mathfrak{J}$ has invariants

$$\frac{\mathfrak{a}_i}{(\mathfrak{n}, \mathfrak{a}_i)} \quad (i = 1, 2, \ldots). \tag{117}$$

Thus $\mathrm{St.}(\mathfrak{n}\mathfrak{J}) < k$ holds exactly when $\mathfrak{n}$ is divisible by the $k$-th invariant of $\mathfrak{J}$. The $k$-th invariant may therefore be characterized as the smallest ideal satisfying this condition.

**63. Irreducible types.** A non-zero type is called irreducible if it is not a sum of two non-zero types. Every finite type can be decomposed into irreducibles. Such an irreducible finite type must be primary and one-stage, and conversely every primary one-stage type is irreducible. The decomposition of a finite type into irreducibles is unique. If $\mathfrak{a}_i$ are the invariants of $[\mathfrak{J}]$ and $\mathfrak{a}_{i,\mathfrak{p}}$ is the largest power of $\mathfrak{p}$ contained in $\mathfrak{a}_i$, then

$$[\mathfrak{J}] = \sum_{i,\mathfrak{p}} T_{\mathfrak{a}_{i,\mathfrak{p}}},$$

where only non-unit ideals contribute. Among finite types, exactly the primary one-stage types are irreducible.

**64. Module types.** Two modules, of the same or of different degrees, are isomorphic if and only if they have the same rank and the same class.

Let $\mathfrak{A}, \mathfrak{B}$ be modules, $\mathrm{Rg.}\,\mathfrak{A} = r$, $\mathrm{Cl.}\,\mathfrak{A} = x$. Choose numbers $c, c'$ whose greatest common divisor belongs to the class $x$, and choose a normal basis $\sigma_1, \ldots, \sigma_r, \sigma_r'$ of $\mathfrak{A}$ such that

$$c\sigma_r + c'\sigma_r' = 0.$$

This is the only relation among those basis rows. Under an isomorphism the row

$$x_1\sigma_1 + \cdots + x_r\sigma_r + x_r'\sigma_r' \tag{118}$$

corresponds to

$$x_1\tau_1 + \cdots + x_r\tau_r + x_r'\tau_r', \tag{119}$$

and equality of rank and class follows. Conversely, if rank and class agree, such bases can be chosen and the correspondence (118)–(119) is an isomorphism.

Thus each natural number $r$ and ideal class $x$ determine a module type, denoted $U_{r,x}$, and for $r = 1$ also $U_x$. If $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$ are ideals and $\sigma_1, \ldots, \sigma_r$ is a basis of $\mathfrak{E}_r$, then for

$$\mathfrak{M} = (\mathfrak{a}_1\sigma_1, \ldots, \mathfrak{a}_r\sigma_r)$$

one has

$$[\mathfrak{M}] = [\mathfrak{a}_1\sigma_1] + \cdots + [\mathfrak{a}_r\sigma_r] \tag{120}$$

and consequently

$$U_{r,x} = U_{x_1} + \cdots + U_{x_r} \quad (x_1 \cdots x_r = x). \tag{121}$$

Among module types, those of rank 1 and only those are irreducible. Every module type of rank $r$ is the sum of $r$ irreducible module types; $r-1$ of these may be chosen arbitrarily, and the last is then determined. In general,

$$U_{r,x} + U_{s,y} = U_{r+s,xy}.$$

**65. Arbitrary types with finite basis.** Let $\mathfrak{J}$ be any ideal system with finite basis, $\mathfrak{J}'$ the subsystem of elements of non-zero order, $\mathrm{Rg.}(\mathfrak{J}/\mathfrak{J}') = r$, and $\mathrm{St.}\,\mathfrak{J}' = s$. If $\mathfrak{J}$ is isomorphic to $\mathfrak{E}_n/\mathfrak{M}$, then the first $n-r$ elementary divisors of $\mathfrak{M}$ are non-zero. Put

$$\mathfrak{a}_i = \mathfrak{e}_i \quad (i = 1, \ldots, n-r), \qquad \mathfrak{a}_i = 1 \quad (i > n-r).$$

These give the invariants of $\mathfrak{J}'$, and therefore

$$n \geq r + s. \tag{122}$$

Let $x = \mathrm{Cl.}(\mathfrak{J}/\mathfrak{J}')$ be the class of the module type and put $\mathfrak{d} = \mathfrak{a}_1 \cdots \mathfrak{a}_s$. Then the type of $\mathfrak{J}$ is determined by

$$r, \quad x, \quad \mathfrak{a}_1, \ldots, \mathfrak{a}_s, \tag{123}$$

where the $\mathfrak{a}_i$ are the invariants of its finite part. Applied to residue-class systems of modules, this says: two such systems are isomorphic if and only if the elementary divisors agree up to units, the difference between degree and rank is the same, and the module classes agree. For modules of the same degree this reduces to equality of elementary divisors and class.

Each basis of $\mathfrak{J}$ gives a module whose residue-class system is isomorphic to $\mathfrak{J}$. From (122), a basis has at least $r+s$ elements. For $s > 0$ there is an irreducible basis with exactly $r+s$ elements. Choose a basis $\alpha_1, \ldots, \alpha_s$ of $\mathfrak{J}'$ with orders $\mathfrak{a}_1, \ldots, \mathfrak{a}_s$, and a basis

$$\rho_1, \ldots, \rho_r, \rho_r'$$

of a subsystem $\mathfrak{T}$ isomorphic to $\mathfrak{J}/\mathfrak{J}'$ with relation $c\rho_r + c'\rho_r' = 0$. Then

$$\alpha_1, \ldots, \alpha_s, \rho_1, \ldots, \rho_r, \rho_r' \tag{124}$$

is a basis. For an element

$$a\rho_r + a'\rho_r' + m\alpha_1, \tag{125}$$

if $\ell$ solves $c\ell = a - m \pmod{\mathfrak{a}_1}$, the same element can be written

$$a(\rho_r + \alpha_1) + a'\rho_r'.$$

Thus in (124) the two elements $\rho_r$ and $\alpha_1$ may be replaced by the single element $\rho_r + \alpha_1$, giving a basis of $r+s$ elements.

## §9. Supplements

**66. Unimodular systems and equivalence.** A unimodular system is a square system of integers of $K$ whose determinant is a unit. Unimodular systems of the same degree reproduce under multiplication, and the reciprocal system is again unimodular.

If $A$ is an $m$-rowed rectangle of degree $n$, $E$ a unimodular system of degree $m$, and $EA = B$, then $A$ and $B$ are left-equivalent. Conversely:

*If $A$ and $B$ are left-equivalent rectangles with the same number of rows, then $B = EA$ with $E$ unimodular.*

More generally: if $A$ and $B$ are left-equivalent $m$-rowed rectangles of rank $r$, if $m > r$, and if $g$ is any integer of $K$, then there is a relation $B = GA$ in which $\det G = g$. Start with

$$B = SA, \tag{126}$$

and, writing the rows of $A, B, S$ as $\alpha_i, \beta_i, \sigma_i$, put

$$\beta_i = \sigma_i A \quad (i = 1, \ldots, m). \tag{127}$$

For $m = r + 1$, the rows $z$ satisfying

$$zA = 0 \tag{128}$$

form a principal module $\mathfrak{T}$ of degree $m$ and rank 1. From basis rows of Md. $S$ together with $\mathfrak{T}$ one obtains a rectangle $T$ such that

$$\mathrm{Md.}\, T = \mathfrak{E}_m. \tag{129}$$

If $z, z'$ are a basis of $\mathfrak{T}$, if $\eta_{ik}$ are the cofactors of the entries $c_{ik}$ of $S$, and $d = \det S$, set

$$u_i = \sum_k \eta_{ki} z_k, \qquad u'_i = \sum_k \eta_{ki} z'_k \quad (i = 1, \ldots, m). \tag{130}$$

Equation (129) gives

$$(u_1, \ldots, u_m, u'_1, \ldots, u'_m) = 1. \tag{131}$$

Hence numbers $q_i, q'_i$ may be chosen so that $\sum_i (q_i u_i + q'_i u'_i) = g$. Replacing the $i$-th row of $S$ by

$$\sigma_i - q_i z + q'_i z'$$

yields a system $G$ with determinant $g$ and $GA = B$.

If $m > r + 1$, use a normal basis of the solution module of $zA = 0$ and unimodular systems $E, F$ to reduce to

$$EA = C, \qquad FB = D, \tag{132}$$

where the last $m - r - 1$ rows vanish. For the rectangles $C', D'$ formed from the first $r + 1$ rows, the already proved case gives a system $G'$ with

$$G'C' = D'. \tag{133}$$

Extending $G'$ by ones on the diagonal gives $(F^{-1}GE)A = B$, with determinant $g$. This proves the theorem. The right-equivalence statements are analogous. If $A$ and $B$ are equivalent rectangles with the same numbers of rows and columns, then $B = PAQ$ with square unimodular $P, Q$.

**67. Extensions.** The restriction to finite algebraic number fields may be removed. If $K$ is any field generated by algebraic numbers, the criteria for equivalence and divisibility of rectangular systems remain the same. Only the definition of module must explicitly require a finite basis. Accordingly, an ideal is any ideal subsystem of the integers of $K$ with finite basis. The division into ideal classes follows the same principle as for finite fields, although the number of classes need not be finite. If $K$ is the field of all algebraic numbers, the number of ideal classes is 1. In such cases, divisibility of a rectangle $B$ by a rectangle $A$ is necessary and sufficient exactly when every elementary divisor of $B$ is divisible by the corresponding elementary divisor of $A$.

**68. Earlier literature.** The oldest investigations related to divisibility of rectangular systems go back to Stephen Smith. In *On Systems of Linear Indeterminate Equations and Congruences*, London Philosophical Transactions 151 (1861), pp. 293–326, the concepts later called determinant-divisors and elementary divisors first appear for rectangular systems with integral entries in the rational number field; the proof that each elementary divisor divides the next is already present there. In 1873 Smith proved that equality of elementary divisors is not only necessary but also sufficient for equivalence of rectangular systems. The divisibility conditions were later completely settled by Frobenius.

The investigations in the section on ideal systems are, in the rational case, only formally different from the theory of groups with commuting elements, developed by Kronecker and by Frobenius–Stickelberger. The connection with module theory is already present there. In algebraic fields, the present paper appears to be the first to carry out the corresponding investigation. The importance of this extension is also shown by I. Schur's recent work on linear substitution groups, which uses results from the first part of the present paper.

Breslau, 10 December 1911.